

HDOC Day-26

C Programming Assignment — Right-aligned Patterns

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Program	B.Tech CSE

Objective

Write a menu-driven C program using a switch statement to print the selected right-aligned pattern. Use nested loops without arrays. Each pattern is implemented as a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu containing the three right-aligned pattern choices.
3. Read the user's choice and the number of rows.
4. Use a switch statement to call the selected pattern function.
5. Use nested loops to print spaces and pattern elements in the required positions.
6. Display the output and stop.

C Program

The following program provides all three activities through a switch-based menu.

```
#include <stdio.h>

int n;

void rightAlignedIncreasingTriangle()
{
    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n - i; j++)
            printf(" ");

        for (int j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

void rightAlignedIncreasingNumberTriangle()
{
    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n - i; j++)
            printf(" ");

        for (int j = 1; j <= i; j++)
            printf("%d ", j);

        printf("\n");
    }
}

void rightAlignedDecreasingTriangle()
{
    for (int i = n; i >= 1; i--)
    {
        for (int j = 1; j <= n - i; j++)
            printf(" ");

        for (int j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

int main()
{
    int choice;

    printf("HDOC Day-26\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter number of rows: ");
    scanf("%d", &n);

    switch (choice)
    {
        case 1:
            rightAlignedIncreasingTriangle();
            break;

        case 2:
            rightAlignedIncreasingNumberTriangle();
            break;

        case 3:
```

```
        rightAlignedDecreasingTriangle();  
        break;  
  
    default:  
        printf("Invalid choice.\n");  
}  
  
return 0;  
}
```

1. Right-aligned increasing triangle

Sample input:

Enter number of rows: 4

Output:

```

      *
     * *
    * * *
   * * * *
```

2. Right-aligned increasing number triangle

Sample input:

Enter number of rows: 4

Output:

```

      1
     1 2
    1 2 3
   1 2 3 4
```

3. Right-aligned decreasing triangle

Sample input:

Enter number of rows: 4

Output:

```

* * * *
 * * *
  * *
   *
```

Test-Case Table

TC	Choice	Input	Expected output / result
1	1	n = 4	Right-aligned increasing triangle of 1–4 stars
2	2	n = 4	Right-aligned rows: 1; 1 2; 1 2 3; 1 2 3 4
3	3	n = 4	Right-aligned decreasing triangle with 4, 3, 2 and 1 stars
4	9	—	Invalid choice message

Conclusion

The program successfully uses a switch statement and nested loops to generate the required right-aligned increasing, number, and decreasing patterns without arrays.

Program: B.Tech CSE **Batch:** 2026–2030